H2O.ai University

H2O Driverless AI Starter Course

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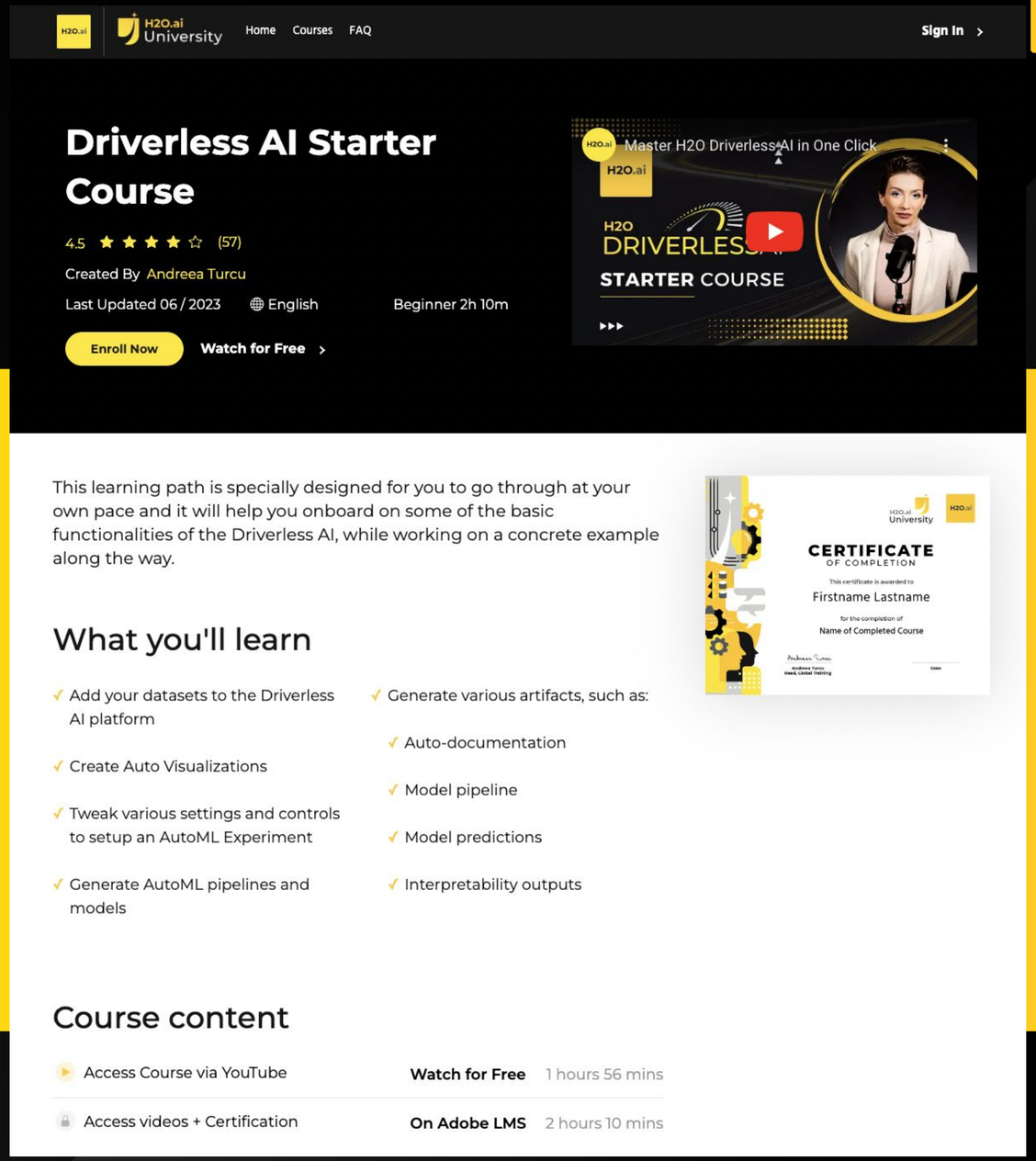
H2O.ai University Website

On our [H2O.ai University]() website you can find:

- An Overview of the [Driverless AI Starter Course]() and
- Access hyperlink to the YouTube [Driverless AI Starter Course]() Playlist.



What you will learn in this course

What is Driverless AI (DAI)?

How to navigate the DAI User Interface

Import, Visualize and Prepare your data

Explore and Understand the Experiment Page

View and Interpret the Completed Experiment

+ ***You can also follow along and complete the assignments.***

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Data Science Project Lifecycle

Data Collection → Feature Engineering → Model Training → Model Evaluation → Model Operations

Gather and prepare data. This is the first step in any machine learning project. No matter what form it comes in, data quality and completeness is a key component to machine learning. Garbage (data) in = garbage (data) out.

Design transformations of predictor variables that may be helpful in modeling. Examples include Box-cox transformations, higher-degree polynomials of existing variables, or simply creating something like "day of week" column from a timestamp.

Use an algorithm to fit a machine learning model. Experiment with different families of algorithms that may be appropriate for the problem at hand. Tune model hyperparameters and combine models that perform well.

Decide on success criteria and measure model performance on data it hasn't seen. Develop evaluation strategies and use cross-validation to measure performance on a sample of data the model hasn't seen before.

Deploy your machine learning model. Take your model and host it in a dedicated environment so that it can make predictions or otherwise score additional data. Monitor for possible "drift" between the original training and new data

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How Driverless AI Works

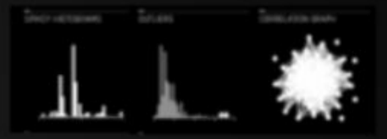
1 Drag and Drop Data

Bring data in from cloud, big data and desktop systems

- HDFS
- SQL
- Local
- Amazon S3
- Snowflake
- Google BigQuery
- Azure Blog Storage

2 Automatic Visualization

Understand the data shape, outliers, missing values, etc.



3 Automatic Model Optimization

Use best practice model recipes and the power of high performance computing to iterate across **thousands of possible models** including advanced feature engineering and parameter tuning

4 Bring Your Own Recipes

Transformations
...

Algorithms
...

Scorers
...

5 Automatic Scoring Pipelines

Deploy ultra-low latency Python or Java Automatic Scoring Pipelines that include feature transformations and models

Automatic Model Optimization

Model Recipes

- i.i.d. data
- Time-series
- NLP
- Image
- ...

Survival of the Fittest

Advanced Feature Engineering + **Algorithm** + **Model Tuning**

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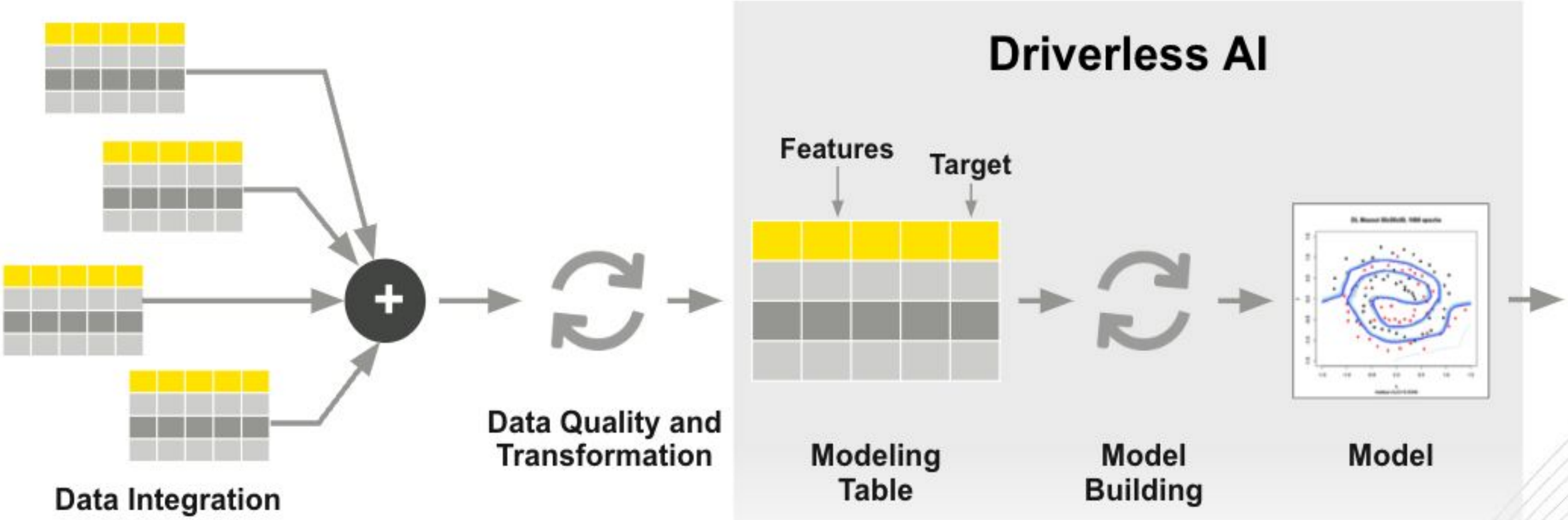
Driverless AI:
Automates Data Science and ML Workflows



Key Capabilities of H2O Driverless AI

- Automatic Visualization
- Automatic Feature Engineering
- Automatic Scoring Pipelines
- Natural Language Processing
- Time Series Forecasting
- Machine Learning Interpretability (MLI)
- Flexibility of Data & Deployment
- Bring-Your-Own Recipes

Best Practices

Data science experts and Kaggle Grandmasters built the tips and techniques within the platform.

Months to Hours

Technology that accelerates the data science and application development workflow.

Explainable AI

Built-in interpretability, fairness, reason codes, visualization, and automated documentation.

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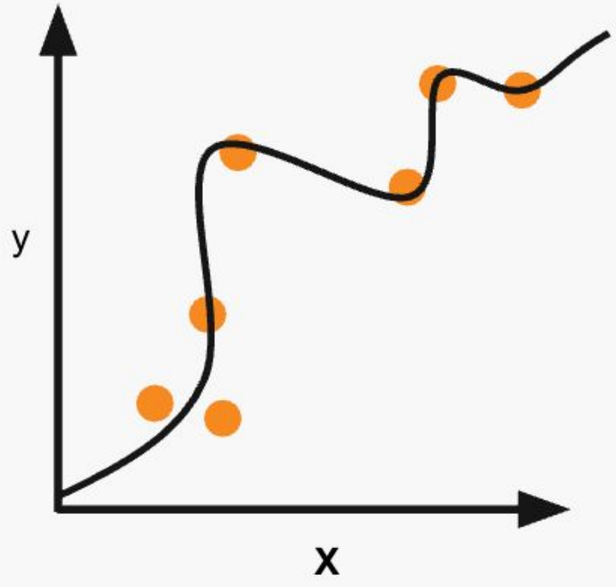
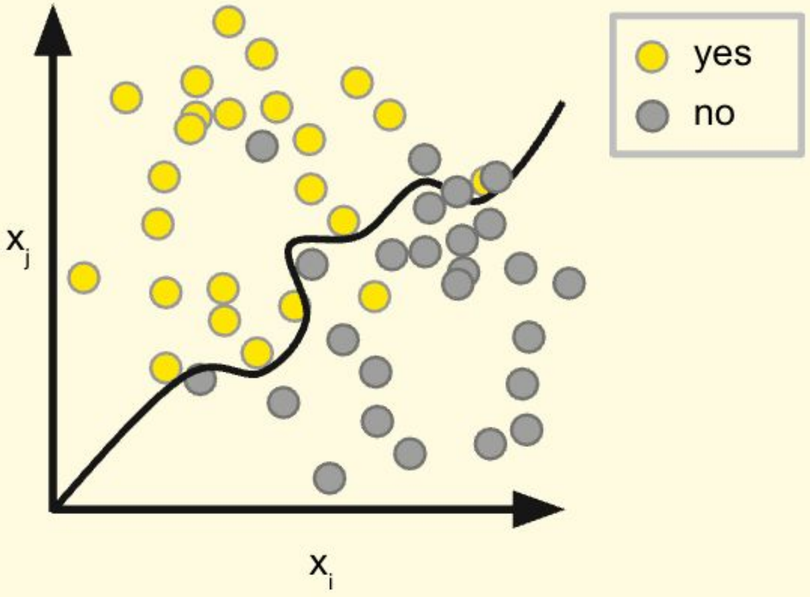
Supervised Learning

Regression:
How much will a customers spend?



Classification:
Will a customer make a purchase? Yes or No



Assignments:

- Assignment 1: Modify recipe using live code
- Assignment 2: Create a new visualization
- Assignment 3: Take a moment to explore the graphs
- Assignment 4: Take the interactive tour
- Assignment 5: Create a new experiment with customized knobs and avgmcc
- Assignment 6: Explore the Shapley Values output
- Assignment 7: Rerun the Baseline experiment
- Assignment 8: Explore the Autodoc

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Assignment 1

Modify Recipe using the live code

Let's create a new dataset by using the **Modify By Recipe** button and selecting the **Live Code** option.

In your newly created dataset, you will:

- Keep the first 7 columns
- Keep the first 5 rows

The new dataset dataset should contain only 5 rows and 7 columns and the **Dataset Details** page should look like the image from the right.

Name	ID	Customer_ID	Name	Age	Occupation	Annual_Income	Monthly_Inhand_Salary
Feature Props	Auto-detect	Auto-detect	Auto-detect	Auto-detect	Auto-detect	Auto-detect	Auto-detect
Storage Type	int	int	str	int	str	real	real
Count	5	5	5	5	5	5	5
Missing	0	0	0	0	0	0	0
Mean	84282.2	13058.2	--	22	--	33170.186	2970.1822
Min	5638	3392	--	20	--	19114.12	1824.8433
Max	103950	27312	--	23	--	38007.66	3320.305
Stdev	43963.4445	8712.3704	--	1.2247	--	7894.6442	641.3222
Freq	1	3	3	2	3	3	3
Unique	5	3	3	3	3	3	3

**You can find a couple of hints on the next slide.*

Assignment 1

Modify Recipe using the live code

Hints:

- You can also use the **Get Preview** option to see the final output of your new dataset based on the Python code.
- Please don't forget to click on the **Apply** button in order to implement the modifications.

Name	ID	Customer_ID	Name	Age	Occupation	Annual_Income	Monthly_Inhand_Salary
Feature Props	Auto-detect	Auto-detect	Auto-detect	Auto-detect	Auto-detect	Auto-detect	Auto-detect
Storage Type	int	int	str	int	str	real	real
Count	5	5	5	5	5	5	5
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Freq	1	3	3	2	3	3	3
Unique	5	3	3	3	3	3	3

**You can find the suggested solution on the next slide.*

Assignment 1

Modify Recipe using the live code

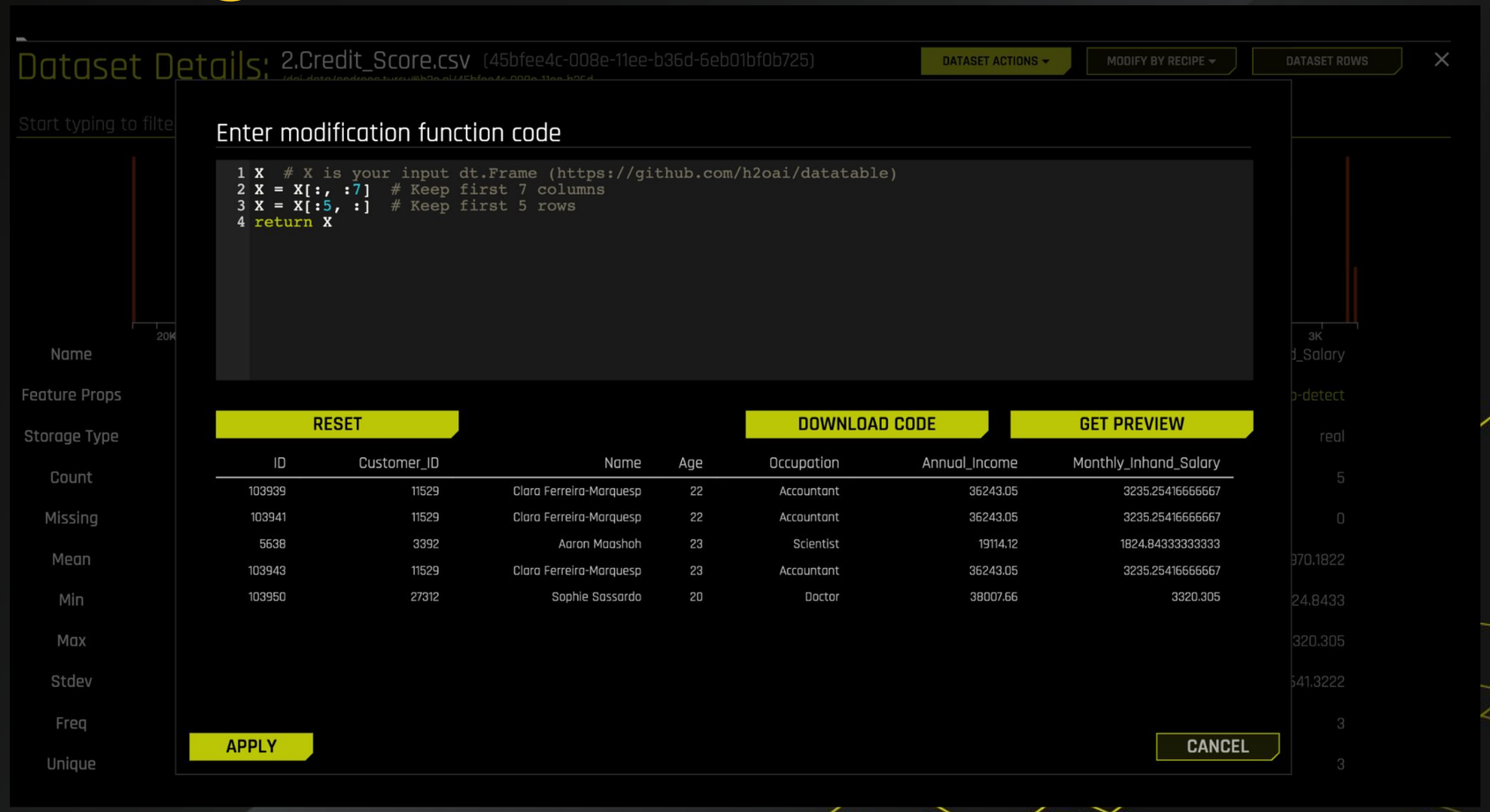
Suggested solution:

X # X is your input dt.Frame (https://github.com/h2oai/datatable)

X = X[:, :7] # Keep first 7 columns

X = X[:5, :] # Keep first 5 rows

return X



Assignment 2
Create a new visualisation

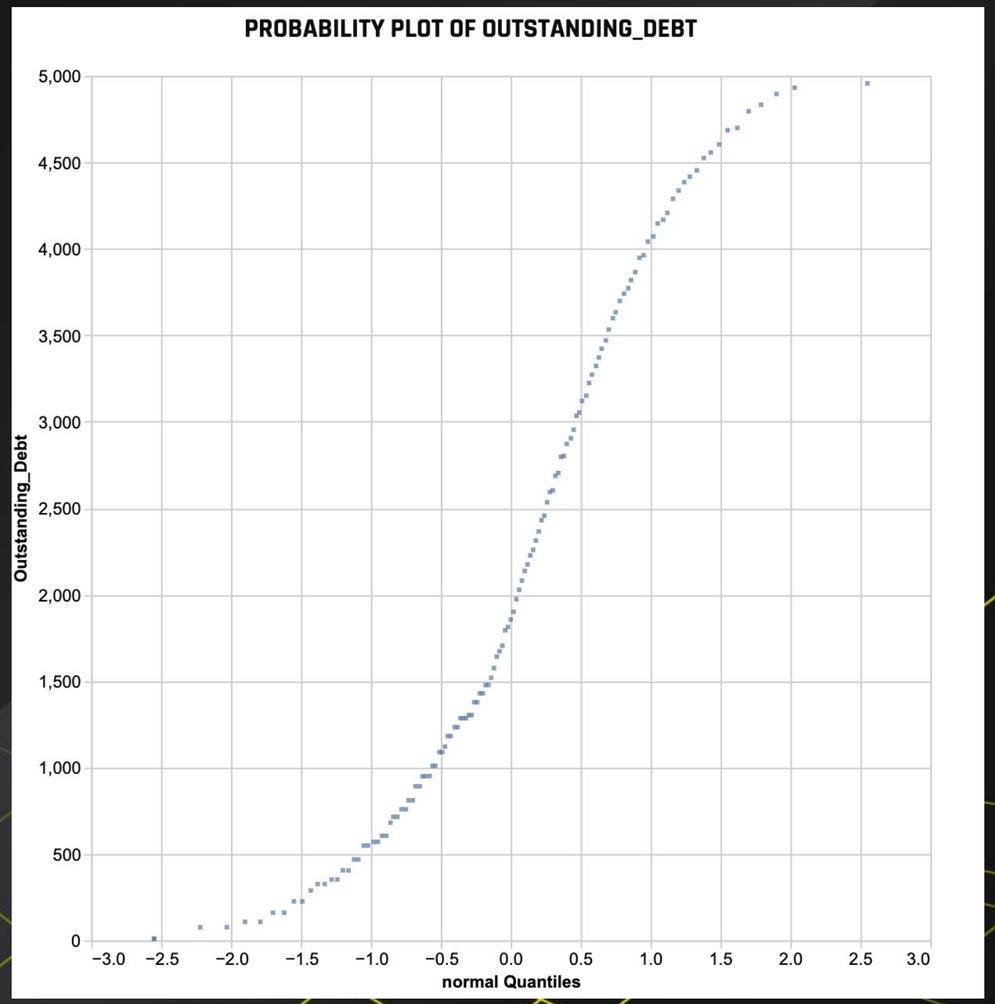
Let's **Add a new graph** on the **Credit_Score.csv** dataset from Driverless AI, by using the **Visualize** action button.

In your new graph, you will create a custom plot where:

- You will need to select the **Probability Plot** type
- You will also choose the **Outstanding_Debt** feature as the Variable Name
- The distribution will be **Normal**
- The mark will be **Square** and
- The transpose option will be **Enabled**

The **Visual** should look like the image from the right.

**You can find the solution on the next slide.*



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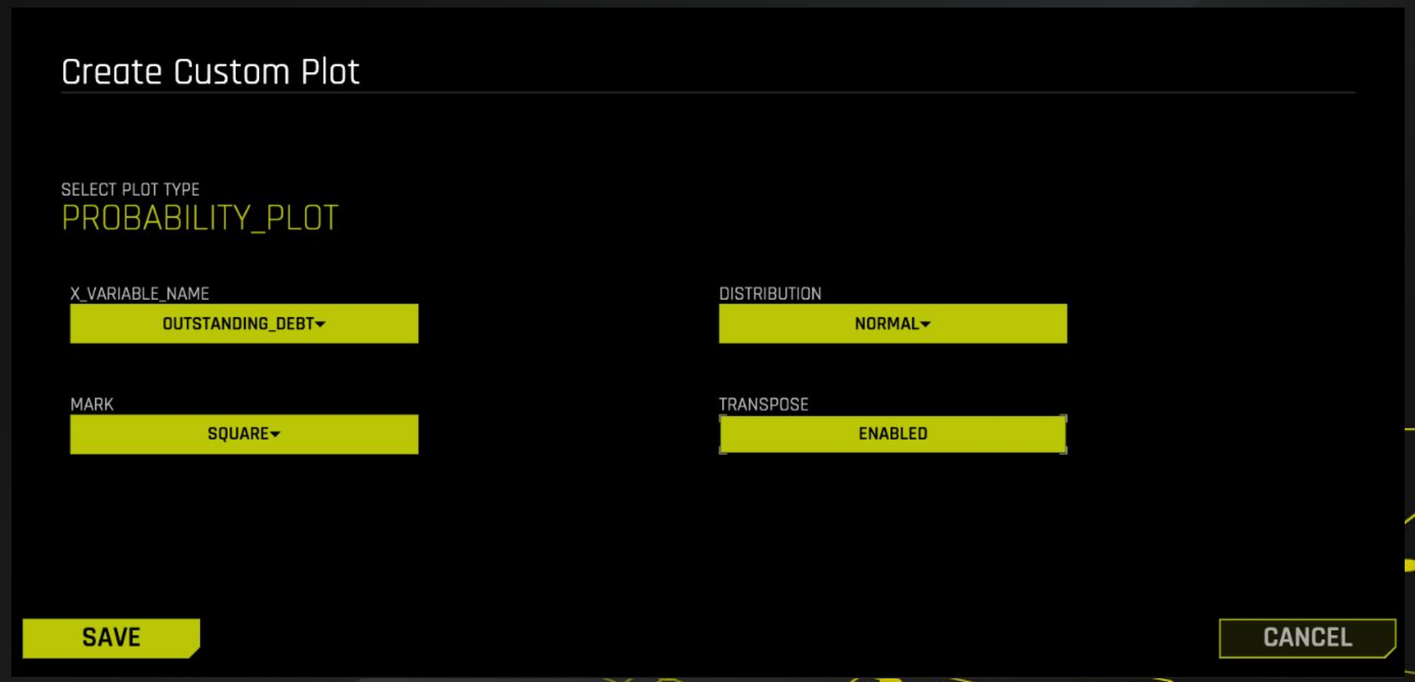
Assignment 2
Create a new visualisation

We can observe from the graph that the **Outstanding_Debt** feature does not follow a normal distribution, as deviations from a straight line suggest departures from normality.

Depending on the business use case, we might want to perform additional transformations on the feature and this can also be done using the Live Code from the previous assignment.



*Feel free to take a couple of minutes and explore on your own the **Add Graph** options*

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Assignment 3

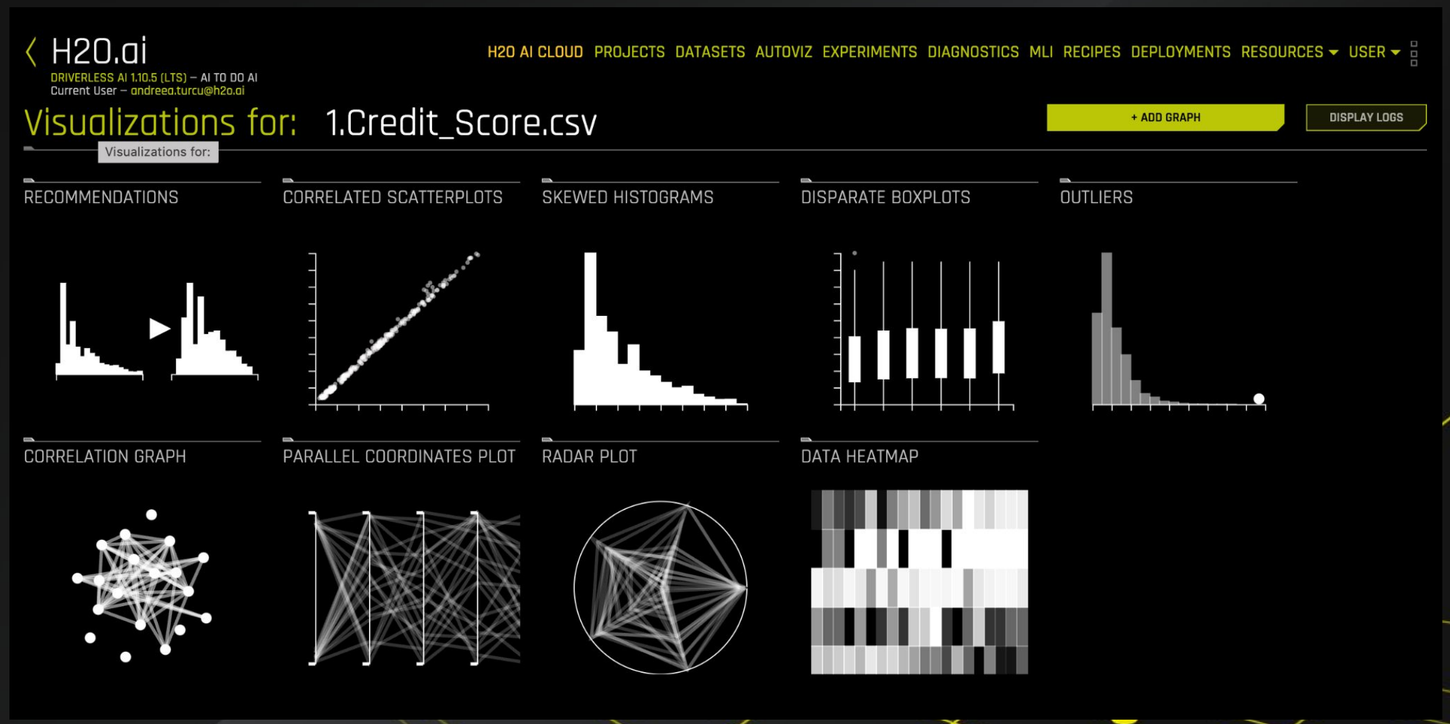
Take a moment to explore the graphs

Feel free to explore on your own and understand:

- the available visualisations for our **Credit_Score** dataset, as well as
- the **Add Graph** options, if not already done.

You have 10 minutes before switching to the next module.

Make the most out of your time!

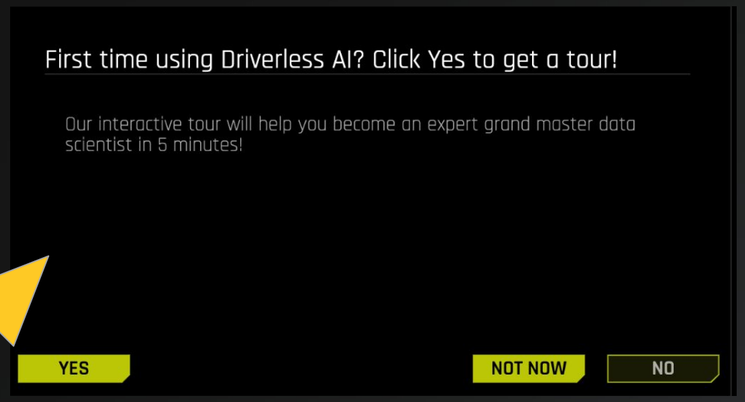


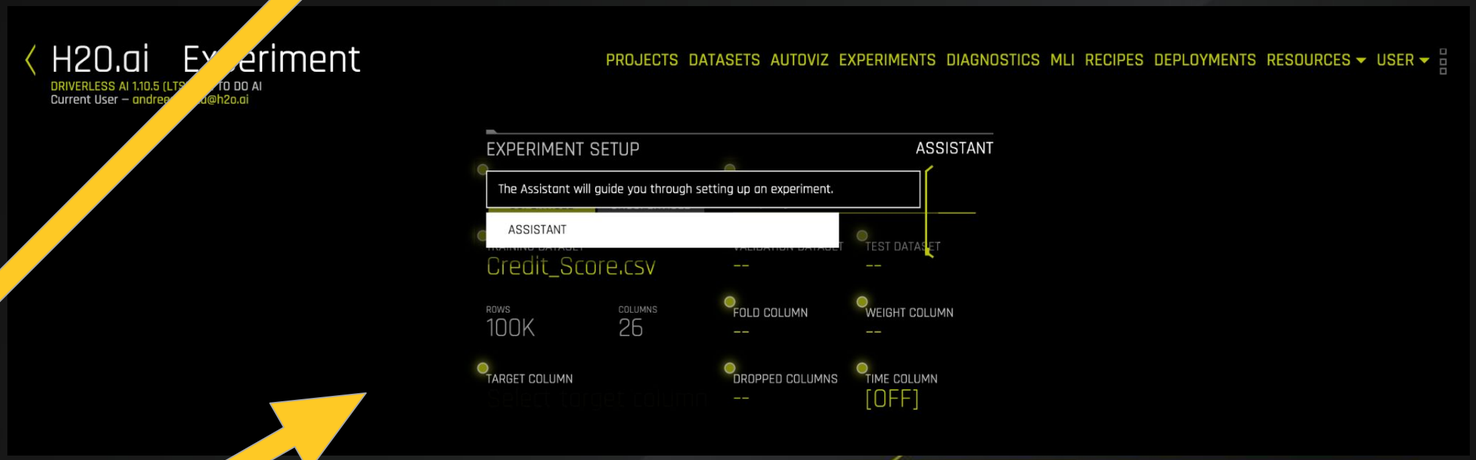
Assignment 4

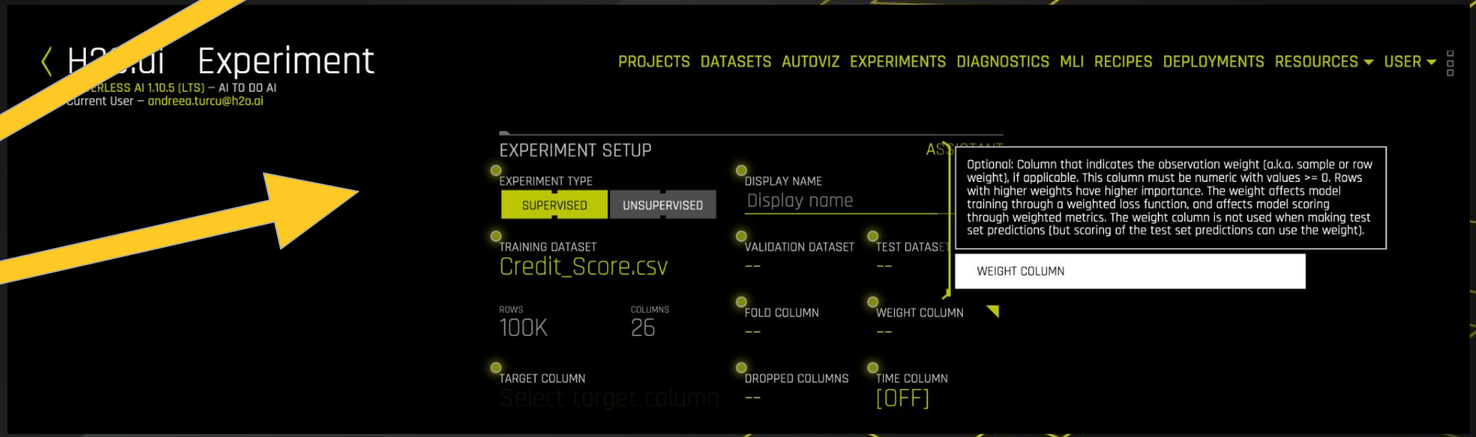
Take the interactive tour

Feel free to explore on your own the **Predict** action button of the **Credit_Score.csv** dataset and:

- take the interactive tour, if not already done, which will help you become an expert grand master data scientist in 5 minutes, or
- activate the Assistant button which will guide you through setting up an experiment
- additionally hover over each one of the Predict page buttons to learn more information about them







Assignment 5
Create a new experiment

Let's create a new experiment from the **Datasets** tab, where we respect exactly the same settings as before, except that this time:

- You will change the knobs and the scorer to the following values:

- Do the settings from the left side change significantly?
- What about the Estimated runtime?

**Go to the next slide when done, to continue the assignment.*

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Assignment 5
Create a new experiment

Next, up, please click on the **Launch Experiment** button, wait for the results and then compare the output of the new experiment to the Baseline one from the **Experiments** tab.

- What do you notice?
- Is the score better or worse for the Baseline?

Let's now click on the **Compare experiments** button from the **Experiments** tab and explore the output even more.

- What anything else do you observe there?

Optional challenge:

Next up, feel free to create additional experiments and tweak the settings as you consider, in order to improve the the score even more.

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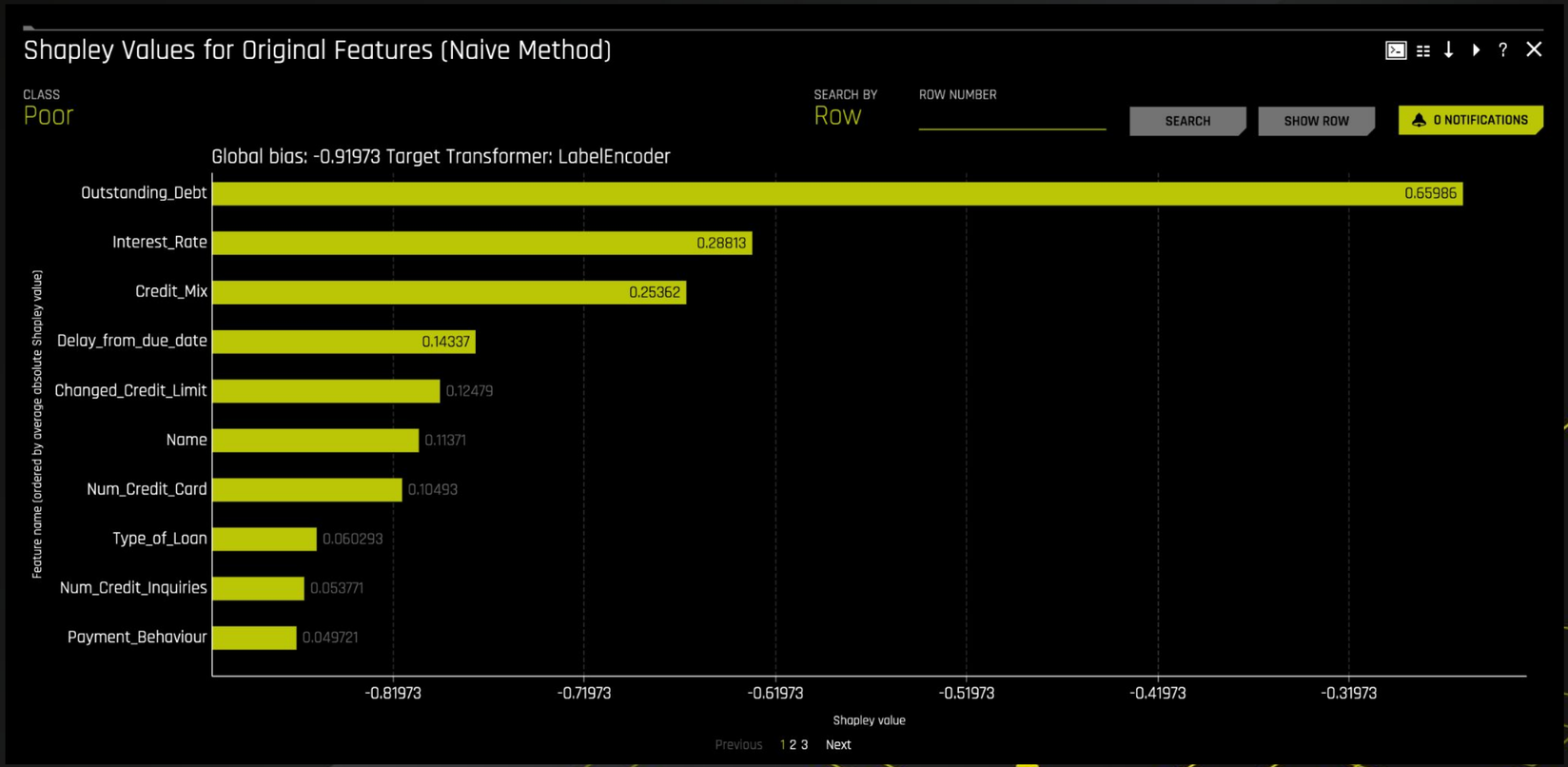
Assignment 6

Explore the Shapley Values output

Please take a look at the **Shapley Values for Original Features (Naive Method)** output from the **MLI** tab and explore the 3 pages underneath the graph.

*Note: The output values might differ from yours, as in this example the **Random Seed** value option before launching the experiment has been skipped.*



**Go to the next slide when done, to continue the assignment.*

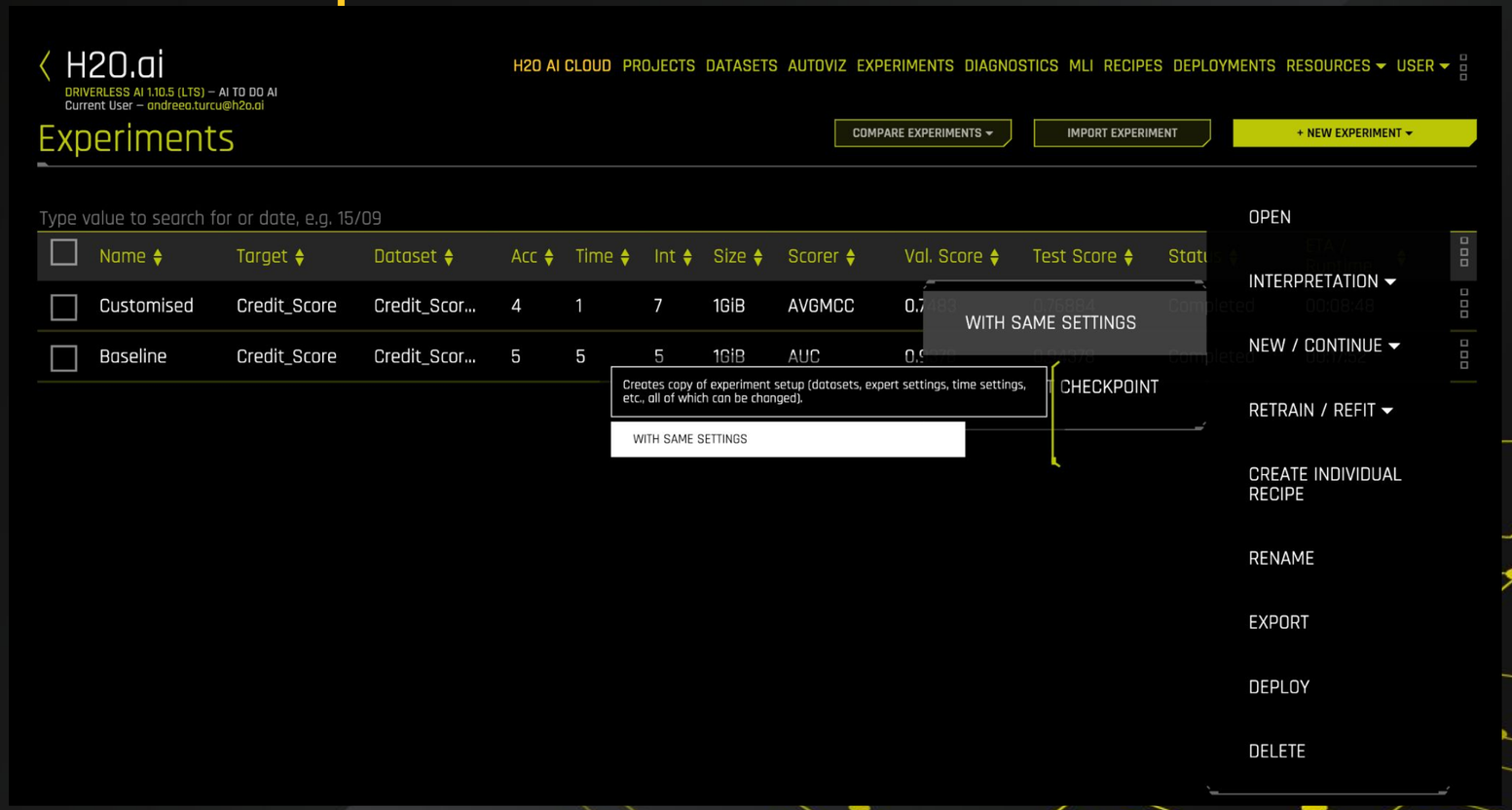
Assignment 7

Rerun the Baseline experiment

Let's rerun the **Baseline** experiment from the **Experiments** tab with the following features dropped:

- **Name** and
- **Customer_ID**

First, click on the action button **New/Continue** for the **Baseline** and select the option **With same settings**.



**Go to the next slide when done, to continue the assignment.*

Assignment 7

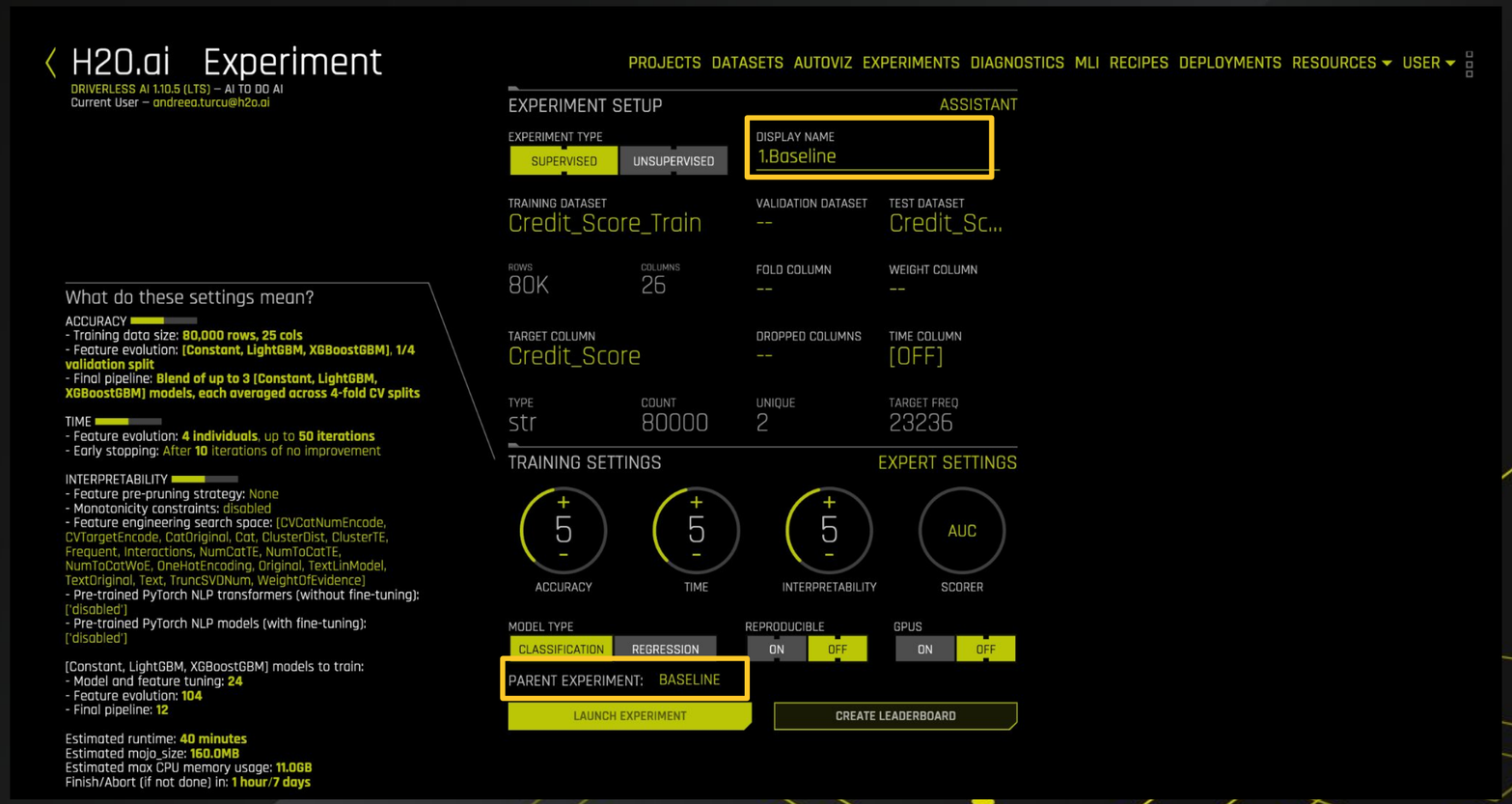
Rerun the Baseline experiment

There will be a new **Experiment** page opening, as on the right side, which has the

Display Name -> 1.Baseline, and the mention:

Parent experiment: Baseline.

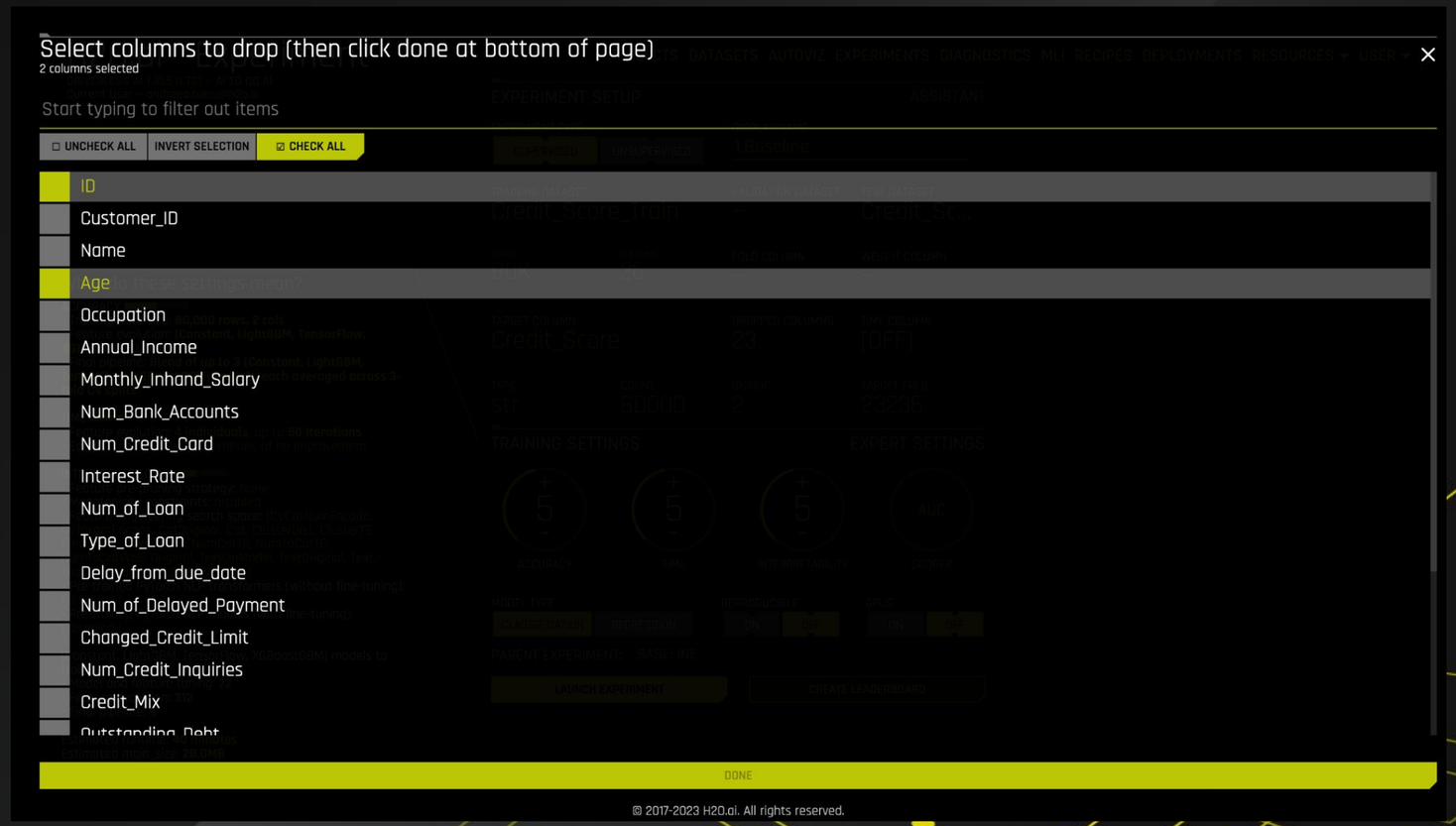
As this new experiment is based on the same initial settings as the **Baseline** experiment, Driverless AI considered the the **1.Baseline** is its child.



**Go to the next slide when done, to continue the assignment.*

Assignment 7

Rerun the Baseline experiment



Nevertheless, in a child experiment you are free to change the settings, so feel free to click the **Dropped Columns** option and select the following features:

- **Name** and
- **Customer_ID**

Then click the **Done** button.

**Go to the next slide when done, to continue the assignment.*

Assignment 7

Rerun the Baseline experiment

Please take note that after clicking the **Done** button Driverless AI takes the two dropped columns in consideration.

Next up, click the **Launch Experiment** button to create a new experiment.

Wait for a couple of minutes and go to the **Experiments** tab once again, to take a look at the output.

- Did the **1.Baseline** score improve as compared to the initial **Baseline**?

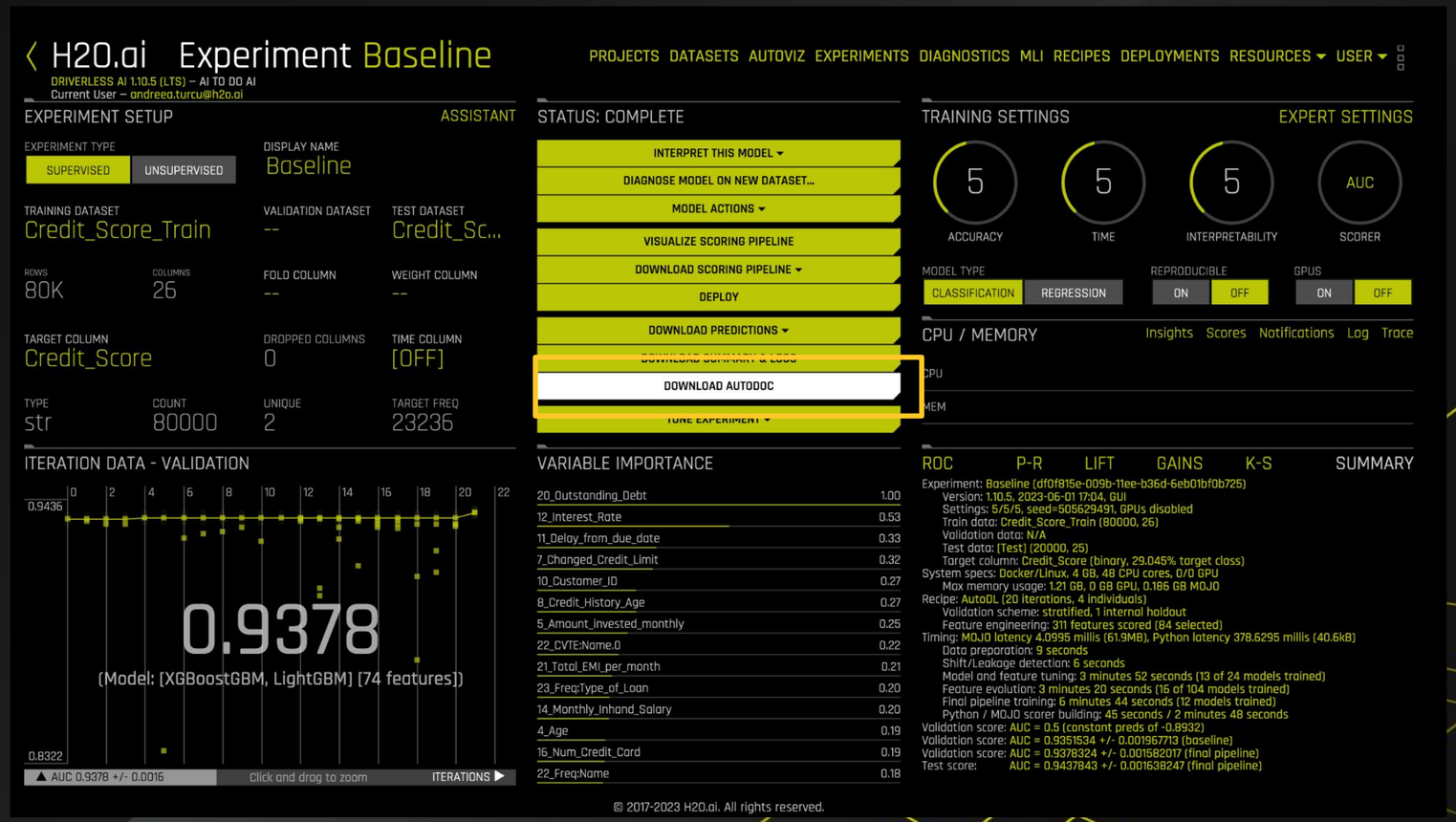
Assignment 8

Explore the Autodoc

Let's download the Autodoc for the **Baseline** model, if not done already.

Take 5 minutes to explore on your own the documentation and then try to answer to the following questions:

- What is the name of the **Final Model** and its type?
- Take a look at the subchapter **Performance of the Final model** and see what are the scorers that have the highest value in the **Performance Table** from below.



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Thank you!